



CI13311-a USB-HS Core family Release Notes

Doc.Revision: 1.1.A

Approval: Miguel Sampaio

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		Title: CI13311-a USB HS Core family Release Notes				Revision: 1.1.A
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Revision History

Doc. Revision	Date	Prepared	Description
0.1	2006-01-16	Joana Braga	• Migrated to Template TP02_DEP_GP
1.0.A	2006-05-10	António Pacheco	• Add section special notes for simulation section • Updated the synthesis procedure section • Updated the simulation Quick start section • Updated supported platforms section • Added latest known bugs section • Updated the dir structure section
1.0.B	2006-07-05	António Pacheco	• Corrected minor grammatical error in section “Bug fixes, new patches” • References to non-supported Solaris and windows platforms were removed • Added the new patches description in section “Revision 1.0.B – 200607”
1.1.A	2006-12-21	Antonio Costa	• Updated Gate count table • Updated Release Notes section • Updated Supported Platforms • Updated Maximum Target Synthesis Frequency

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1 Performance Summary

1.1 Core Summary

1.1.1 Performance

480Mbps High speed serial transfer rate.

1.1.2 Maximum Target Synthesis Frequency

These results were obtained using different TSMC process target as a reference.

	TMSC 0.18um	TSMC 0.13um	TMSC 90nm
Maximum Target Synthesis Frequency	290MHz	367MHz	450MHz

1.1.3 Core Area at Minimum Target Synthesis

50MHz in Artisan TSMC 90nm Standard Cell CMOS Library.

Clock configuration 0 is used for all. All gate count numbers are for the BVCI bus interface.

	Base	Add per Endpoint	Min. (2 Bidir EP)	Max. (16 bidir EP)
Device (UTMI Phy Interface)	19.6k	2.3k	24.2k	56.6k
Device (ULPI Interface)	20.8k	2.3k	25.5k	57.9k

Table 1 - Device Product Gate Count

	Base	Add per Endpoint	Min. (2 Bidir EP)	Max. (16 bidir EP)
OTG (UTMI+ Phy Interface)	30.5k	2.3k	35.1k	67.6k
OTG (ULPI Phy Interface)	31.8k	2.3k	36.4k	68.8k

Table 2 - OTG Product Gate Count

	Base
SPH (UTMI+ Phy Interface)	25.2k
SPH (ULPI Phy Interface)	26.5k

Table 3 - Single Port Host Product Gate Count

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	Base	Add per Port	Min. (2 Ports)	Max. (8 Ports)
MPH (UTMI+ Phy Interface)	40.2k	6.5k	53.2k	92.3k
MPH (ULPI Phy Interface)	41.5k	7.8k	57.1k	103.8k
Increase base gate count when the number of internal periodic Transaction Translator contexts is increased from 4 to 16.				12.9k

Table 4 - Multi-Port Host Product Gate Count

Add to use AMBA-AHB w/Retry & Split Support Interface in place of BVCI.	2.6k
Add to use AMBA-AHB w/Retry & Split Support and APB Slave Interface in place of BVCI.	2.4k
Add for software selectable UTMI/UTMI+ Interface Logic. Applies to baselines above that don't already include a UTMI/UTMI+ Interface (ie. ULPI)	1.1k (per port for MPH)
Add for software selectable ULPI Interface Logic. Applies to baselines above what don't already include a ULPI Interface (ie. UTMI/UTMI+)	2.4k (per port for MPH)
Add for software selectable Serial Interface Engine for FS/LS Signaling.	2.4k (per port for MPH)
Add for Core with clock configuration 1	2.4k
Add for Core with clock configuration 2	1.4k
Add for Core with clock configuration 3	3.9k

Table 5 - Gate Count Modifiers

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2 Simulation & Synthesis Quick Start

The USB-HS product contains a number of simulation and synthesis files that automate the performance of repetitive tasks. The information in this section is provided as a quick start. For more information see the Reference Manual.

2.1.1 Build Scripts (Simulation)

In the simulation/scripts directory, the scripts buildsim.pl can be used to compile the simulation database. A second scripts runtest.pl, which calls buildsim.pl, can be used to compile and start a simulation for one of three simulators:

- Model Technology - Modelsim
- Cadence - NC Verilog
- Synopsys - VCS

See the comments and source code in buildsim.pl and runtest.pl for more information.
The user options for the buildsim.pl & runtest.pl scripts are:

Options	Descriptions
--sim mti	Calls the buildsim/runtest for use with ModelSim.
--sim nc	Calls the buildsim/runtest for use with NCVerilog. [Verilog Release Only]
--sim vcs	Calls the buildsim/runtest for use with Synopsys VCS. [Verilog Release Only]
-a	Run AMBA™-AHB simulation.
-b	[default] Run BVCi simulation.
-p	Run AMBA™-AHB master + AMBA™ APB 3.0 for slave interface

Note: [-a, -b and -p switches are mutually exclusive]

Table 6 - buildsim.pl script options (sim directory)

2.1.2 Build Scripts (Synthesis)

To run synthesis using three scripts are supplied in the folder tool_data / design_compiler / scripts, one is for bvci option, the other is for amba option and the third is for APB option.

To run the file, issue the following command with the appropriate synthesis environment:

```
dc_shell-t -f scripts/<script_file.tcl>
```

2.1.3 Simulation Quick Start

NOTE: PERL 5.6 OR LATER IS REQUIRED FOR THE SIMULATION QUICK START PROCEDURE. THE PERL BINARY MUST BE IN THE USER'S PATH.

2.1.3.1 ModelSim (VHDL and Verilog)

After installing the release, executing the following commands will get ModelSim users simulating the design in a standalone environment.

1. Change the directory to the simulation directory: **cd simulation/scripts**
2. Execute the run test command file: **perl runtest.pl --sim mti** on Linux to run to BVCi test bench or **perl runtest.pl -a --sim mti** on Linux to run to AMBA™-AHB test bench, for slave AMBA-APB and master AMBA-AHB testbench run **perl runtest.pl -p --sim mti**.

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2.1.3.2 NCVerilog (Verilog Only)

1. Change the directory to the simulation directory: **cd simulation/scripts**
2. Execute the run test command file: **perl runtest.pl --sim nc** on Linux {BVCI testbench} OR **perl runtest.pl -a --sim nc** on Linux {AMBA™-AHB testbench}, for slave AMBA-APB and master AMBA-AHB testbench run **perl runtest.pl -p --sim nc**.

2.1.3.3 Synopsys VCS (Verilog Only)

1. Change the directory to the simulation directory: **cd simulation/scripts**
2. Execute the run test command file: **perl runtest.pl --sim vcs** on Linux {BVCI testbench} OR **perl runtest.pl -a --sim vcs** on Linux {AMBA™-AHB testbench}, for slave AMBA-APB and master AMBA-AHB tetbench run **perl runtest.pl -p --sim vcs**

2.1.4 Supported Platforms

The Chipidea USB-HS products supports the following products and platforms:

- | | | |
|------------|---|--------------------|
| • Adobe | Acroread | 4.05 and later |
| • Cadence | NCsim Verilog | 5.7 |
| • Mentor | ModelSim | 6.1c (Linux) |
| • Red Hat | Linux Enterprise | 3.0 |
| • Synopsys | <ul style="list-style-type: none"> ▪ DC ▪ VCS | 2005.09-SP3
7.1 |
| • Xilinx | ISEAlliance | 8.1 |

NOTE The binary plug-in for Modelsim, `SimSocket.dll`, has been found to be compatible with some older versions of MTI. However, if discrepancies arise when attempting to run simulations using a version of MTI other than that shown above, Chipidea recommends using the version shown in the table above.

NOTE The binary plug-in for NCVerilog, `SimSocketNCVerilog.so`, has been found to be compatible with some older versions of NCVerilog. However, if discrepancies arise when attempting to run simulations using a version of NCVerilog other than that shown above, Chipidea recommends using the version shown in the table above.

NOTE The binary library for VCS, `co_simulation_vcs_verilog.a`, has been found to be compatible with some older versions of VCS. However, if discrepancies arise when attempting to run simulations using a version of VCS other than that shown above, Chipidea recommends using the version shown in the table above.

NOTE The binary library for VCS & GCC, `co_simulation_vcs_verilog.a`, is built using g++ and therefore it is necessary to link with the g++ libraries while build the simulator executable. The makefile examples for VCS call out the switches “-cc g++ and -ld g++” on the VCS executable. GCC/G++ can be obtain at <http://www.gnu.org> . The GCC version we use is 3.2.3 20030502.

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3 Release notes

3.1 Revision 1.1.A – 200612

Revision 1.1.A is a major release with some new features and resolving some minor RTL / Testbench issues :

- **New Features (RTL)**
 - Master bus interface extended to support AMBA AHB incremental fixed bursts INCR4/INCR8/INCR16.
 - Slave bus interface now can be selected to be AMBA AHB or AMBA APB 3.0.
- **Enhancements (RTL)**
 - The following file “vusb_hs_rx_buf.vhdl” was modified to be compatible with NCVerilog.
- **Bug Fixes (RTL)**
 - [Host, OTG]: The Host core with FS device directly attached, without HUB, after a ISO IN with CRC error on Data packet, was not recording the error on the Status bit ERR bit of the siTD. (ID1649)
 - [Device, OTG]: The pe_up_dev_rst_irq signal can have a glitch because of the state change in the dc_ophy_port_state_r signal. With the correct timing, the up_clk could capture this glitch and generate a false interrupt that the software would see. (ID1372)
 - [Host, OTG]: qTD overlay information is overwritten in the context registers, but the ERR bit, which is shared with the PING control bit in High Speed, was not updated. (ID1198)
 - [Host, OTG]: Host high bandwidth ISO packet sequence was not as expected for Transfer sizes lower than 2048bytes using Mult=3 (ID1181)
 - [Host, OTG]: The frame interval repeatability, TRFI (difference in frame interval between two successive frames), must be less than 0.5 full-speed bit times. The microframe interval repeatability, THSRFI (difference in the microframe interval between two successive microframes, measured at the host), must be less than 4 high-speed bit times. Each hub may introduce at most 4 additional high-speed bits of microframe jitter. In clock configuration 1& 3 it was possible that depending on what the system/auxiliary clock frequency was, that this could have be violated. The SOF jitter has been removed for Host and OTG, but it still could exist for Multi-Port Host. For Multi-Port Host, the jitter will only be removed if the new configuration constant, VUSB_HS_MULTI_PORT_FREQ_LOCKED, is set to a 1 and all the necessary clocking requirements have been met to set that constant to a 1. (CR576)
 - [Multi-Port Host]: The port numbering scheme which was originally implemented in the Multi Port Host counted the ports from 0 to N. It was discovered that FS/LS devices would not work with this scheme, and that port numbering should count from 1 to N. (CR618)
 - [Device, OTG]: Under certain circumstances, a packet would be sent which was too long (oversized and corrupted) during prime-break. (CR631)
 - [OTG]: The ID pin loop-back when the pull-up was disengaged was not correct. Now, the OTG block will always pass to the OTG ID, status regardless of the ID pull-up. (CR624)
 - [ULPI Host]: In host mode in the ULPI core, a remote wakeup could be interpreted as a disconnect. This issue involved the latency in asserting synchronous mode. In some instances, the host would not properly latch the K state. The core then saw that it woke up with a J. Eventually the host wouldn't takeover the resume, and would then later would see an SE0 and presume a disconnect occurred. The latency in asserting synchronous mode has been fixed, so this is no longer an issue. (CR619)
 - [ULPI Device]: In device mode in the ULPI core, a remote wakeup could be interpreted as a disconnect. In some instances, the host would not properly latch the K state. (CR642)
 - [ULPI]: The registers in the ULPI block were a separate reset domain (xvcr_porst) from the rest of the USB core, which was on the xvcr_rst domain. With the registers on separate resets, there were

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- signals from the opposite reset domain driving the ULPI registers, feeding through multiple layers of logic. This caused unknowns to be propagated in gate-level simulations, down to the ulpi_rx_data and ulpi_tx_data registers. That could lead to corruption of the address and data fields and erroneous ULPI accesses, not only in simulation, but in silicon. All resets are now on the xcvr_rst/xcvr_rst_a domain. (CR629, CR632)
- **Bug Fixes (Testbench)**
 - [Responder]: The Responder was improved to be compatible with a wider range of USB PHY models. (ID1099, ID1100, ID1117)
 - [Test software]: When the OTG Controller switches to device mode, after reconnect, the C++ testcode was expecting to see the bit connect change detect be set. But this is not supposed to happen. (ID1339)
 - [Test software]: The issue was that when a Controller is configured to have 2 Endpoints, the "Device,HandShake, Multi Endpoint*" can not be run. (ID1156)
 - [Test software]: ULPI registers tests were not checking the correct values defined in ULPI specification 1v1. (ID1098)
 - [Test software]: Fixed bug in SimSocket.cpp related to NCVerilog simulation. Changed the names of the shared libraries to "BFM_MTIVHDL.so", "BFM_MTIVerilog.so", "BFM_NCVHDL.so", "BFM_NCVerilog.so", "BFM_VCSVerilog.a" and "BFM_VCSVerilog_legacy.a".
 - [Test software]: Added a config string to insert a 4ms delay before the reset. Some customer have very slow systems require this delay. (CR612).
 - [Test software]: Disabled low power state after port change interrupt. Before the change, OTG simulations without a power controller and a proper wakeup signal (ie. PHY never enters low power) would fail because the low power signal will be set in one suspend test and remain asserted in the next. In the subsequent suspend operation, the high speed blanking interval is skipped due to the low power mode, causing the simulation to fail. (CR623)
 - [Test software]: The GPIO timer would stop when the PHY was in low power mode and the clock would go inactive, which would cause the simulation to hang. The software was changed such that the GPIO timer was not used when low power PHY tests were performed. (CR638)
 - **Bug Fixes (Embedded Testbench)**
 - [Embedded Test software]: With some slower embedded software/processors, the instant connect attempt could be aborted, then the instant connect polling attempt constant MAX_INSTANT_CONNECT_POLL_ATTEMPTS was increased 5 times.(ID1473)
 - [Embedded Test software]: Embedded Test Code was written in a way that all Host-generated Control transfers are set to have a transfer payload of 8 bytes, regardless of how many of those 8 bytes are useful for a specific command. This was ok in the standalone environment because all memory space is initialized, and no Xs are ever read when it accesses the non-initialized data. But in a customer's SoC system, the non-initialized data bytes could be read back as Xs if those RAM locations were not initialized. The corresponding functions were changed, in order to always complete the 8 bytes of payload data with 0s. (ID1474)
 - **Documentation Updates**
 - Please check the reference manual for complete updates history.
 - Added new configuration constant, VUSB_HS_MULTI_PORT_FREQ_LOCKED. This constant is only to be altered for the multi-port host product. Leave 0 for device, OTG, and single-port host. The effect of setting the constant will ensure that the jitter alignment circuit is operating to minimize SOF jitter on each port and ensure compliance to the frame repeatability interval in the USB specification. The documentation on VUSB_HS_MULTI_PORT_FREQ_LOCKED should be fully read before setting it in a multi-port host implementation. (CR576)

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3.2 Revision 1.0.B – 200607

Revision 1.0.B is a minor patch release resolving the following RTL issues:

- **Enhancements (RTL)**
 - [ALL] The RX/TX latency buffers uses dual port RAMS with completely independent read and write ports. This means that a write and a read to the same address at the same time was possible. The DPSRAM from some vendors do not independent Write and Read ports, and if the two port clock domains were asynchronous, this could cause a timing violation during the write on those RAMS. The new patch ensures that a Write and Read to the same location on the same time will not happen. BUG ID #1240.
- **Enhancements (Testbench)**
 - [ALL] Added assertion to [dpsram.v|vhdl] in order the detect the write / read collision and break the simulation on that case. BUG ID #1240.
- **Bug fixes (RTL)**
 - [ALL] Lint issues solved in the following files: vusb_hs_portctrl_sm.[v|vhdl] and vusb_hs_dma_hst_sm.[v|vhdl]. The issue was related with sensibility list and in vusb_hs_dma_hst_sm.v the problem was related to a switched port direction. BUGID #1205.

3.3 Revision 1.0.A – 200605

Revision 1.0.A is the first CI release after the controller assets were acquired. This version corresponds to the last revision (REV4.2.0.200506) before its acquisition by Chipidea with the following list of corrections:

- **Bug Fixes (RTL) new patches**
 - [OTG/SPH/MPH] When in host mode, it was found that the USB error interrupt bit of register USBINTERR is not fired after a IN transaction error. This bug was not silicon critical since the appropriate error bits in the asynchronous / synchronous list are correctly flagged, so the software could workaround this issue on the previous version. BUG ID #990.
 - [OTG/SPH/MPH] When in host mode, according to EHCI spec, when a HS ISO IN transaction fails due to short packet, BTO or other, the transaction must be halted and the controller move to the next ISO iTD on next microframe. Previous release issued the ISO IN 3 times before giving up, instead of dropping it on the first failure, as stated by specification. Although there was no SW workaround the bug was not silicon critical, since in practical terms this bug would result in waste of bandwidth. The current release was fixed to reflect the correct ISO IN behavioral. BUG ID #1084.
 - [OTG/SPH/MPH] When in host mode, during HS resume sequence, the PORTCTRL state machine didn't sense the line-state for SE0 (confirming that line-state is HS idle) before going to IDLE state. Some PHY's can take a few μ S putting HS idle state, and since the state machine didn't wait for the PHY it can send HS signaling to the line, with the line still in FS (in practical terms the device would ignore this). This bug only shows up with some non-Chipidea PHYs. BUG ID #1095.
- **Bug Fixes (RTL) covered by existing patches**
 - [ALL] Improved timing on master read data bus into the core. Timing paths going into the core from the master read data had a path all the way to the latency RAMs. This was as intended, however a register was added in the path, in the dma_mem_arb block, for more favorable timing.
 - [ALL] The AHB master could respond to an error response for another master on systems that use the req/grant protocol. This issue does not affect AMBA Lite implementations, but is silicon critical for systems which use req/grant protocol. This issue has now been fixed.
 - [ALL] Signal ulpi_tx_pkt_state_nxt was missing from the sensitivity list of process ophy_decode_PROC. This signal has been added to the sensitivity list.
 - [ALL] In Host mode in the ULPI core, a remote wakeup could be interpreted as a disconnect. This issue involved the latency in asserting synchronous mode. In some instances, the host would not

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- properly latch the K state. The core then saw that it woke up with a J. Eventually the host wouldn't takeover the resume, and would then later would see an SE0 and presume a disconnect occurred. The latency in asserting synchronous mode has been fixed, so this is no longer an issue.
- [ALL] The registers in the ULPI block were a separate reset domain (xcvr_porst) from the rest of the USB core, which was on the xcvr_rst domain. With the registers on separate resets, there were signals from the opposite reset domain driving the ULPI registers, feeding through multiple layers of logic. This caused unknowns to be propagated in gate-level simulations, down to the ulpi_rx_data and ulpi_tx_data registers. That could lead to corruption of the address and data fields and erroneous ULPI accesses, not only in simulation, but in silicon. All resets are now on the xcvr_rst/xcvr_rst_a domain except for the ulpi_enable_PROC, where its only input also comes from the xcvr_porst domain. Other changes include automatically waking up the PHY should an aborted register write cause the PHY to go into one of the low power modes.
 - [ALL] A combinational loop existed in the logic:
xcvr_ophy_tx_valid_i depends on ulpi_host_suspended_i
ulpi_host_suspended_i depends on ulpi_tx_pkt_state_nxt
ulpi_tx_pkt_state_nxt depends on xcvr_ophy_tx_valid_I
A register was placed in the ulpi_host_suspended_i path to fix this.
 - [DEV/OTG] Under certain circumstances, a packet would be sent which was too long (oversized and corrupted) during prime-break.
 - [MPH] The port numbering scheme which was originally implemented in the Multi Port Host counted the ports from 0 to N. It was discovered that FS/LS devices would not work with this scheme, and it was determined that port numbering should count from 1 to N.

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3.4 Deliverables File List

The following diagram illustrates the directory structure for the USB-HS design after installation is complete.

Folder/File	Description
 CI13410C CI13510C CI13610C CI13710C	Top level of installation
 doc	Online documentation
 simulation	Simulation base dir
 scripts	Simulation launch location using scripts
 testbench	SW test code to run simulations
 bfm	Processor model based test code
 timing	Timings files of the synthesis netlist
 tool_data	Tools' folder
 design_compiler	Scripts and dir structure for synthesis
 modelsim	Files for the modelsim MTI's simulator
 scripts	Misc perl scripts to automate things
 vcs	Synopsys' VCS specific file
 behavioral_v	Behavioral models in verilog
 behavioral_vhdl	Behavioral models in VHDL
 rtl_v	RTL verilog files
 rtl_vhdl	RTL VHDL files
 gatelevel_v	Synthesis verilog netlist
 gatelevel_vhdl	Synthesis VHDL netlist

Table 7 - Directory structure

3.4.1 DOC

This directory has one sub directory containing the following Adobe Acrobat documents:

- USB-HS Core Release Notes
- USB-HS Core Reference (appropriate to this release)
- CI_USBHS_Embedded_Integration_Test_Suite_0v3

3.4.2 SIMULATION

This directory contains files needed for simulation. This directory contains:

- Perl script used to compile source code for simulation
- Perl script to run simulations

3.4.3 TESTBENCH

The testbench directory contains the integration test software. This “C++” source code source and support files are used either on a system processor or on the supplied Bus Functional Model (BFM.)

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3.4.4 RTL_V or RTL_VHDL

This directory contains the VHDL [verilog] source code for the USB-HS design including the files in the table below:

File	OTG / DEV / SPH / MPH	Description
vusb_hs_cfg.vhd1	All	USB-HS Constant Definitions
vusb_hs_core_dev_bvci.vhd1	DEV	USB-HS Top Level File – BVCi
vusb_hs_core_dev_amba_apb.vhd1	DEV	USB-HS Top Level File – AMBA-AHB
vusb_hs_core_otg_bvci.vhd1	OTG	USB-HS Top Level File – BVCi
vusb_hs_core_otg_amba_apb.vhd1	OTG	USB-HS Top Level File – AMBA-AHB
vusb_hs_core_sph_bvci.vhd1	SPH	USB-HS Top Level File – BVCi
vusb_hs_core_sph_amba_apb.vhd1	SPH	USB-HS Top Level File – AMBA-AHB
vusb_hs_core_mph_bvci.vhd1	MPH	USB-HS Top Level File – BVCi
vusb_hs_core_mph_amba_apb.vhd1	MPH	USB-HS Top Level File – AMBA-AHB
vusb_hs_dma_context.vhd1	All	DMA Data Structure Storage
vusb_hs_dma_up_int.vhd1	All	DMA Microprocessor Interface
vusb_hs_dma_traf.vhd1	All	DMA Packet Movement Logic
vusb_hs_dma_traf_tt.vhd1	MPH	DMA Packet Movement Logic - TT
vusb_hs_dma_dev_sm.vhd1	DEV, OTG	DMA Device State Machine
vusb_hs_dma_hst_sm.vhd1	OTG, SPH, MPH	DMA Host State Machine
vusb_hs_dma_mem_arb_bvci.vhd1	All	DMA Bus Interface Logic
vusb_hs_dma_mem_arb_amba.vhd1	All	DMA Bus Interface Logic
vusb_hs_dma_dev_bvci.vhd1	DEV	DMA Device Controller Structure – BVCi
vusb_hs_dma_dev_amba_apb.vhd1	DEV	DMA Device Controller Structure – AMBA-AHB
vusb_hs_dma_otg_bvci.vhd1	OTG	DMA Device Controller Structure – BVCi
vusb_hs_dma_otg_amba_apb.vhd1	OTG	DMA Device Controller Structure – AMBA-AHB
Vusb_hs_dma_sph_bvci.vhd1	SPH	DMA Device Controller Structure – BVCi
Vusb_hs_dma_sph_amba_apb.vhd1	SPH	DMA Device Controller Structure – AMBA-AHB
Vusb_hs_dma_mph_bvci.vhd1	MPH	DMA Device Controller

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File	OTG / DEV / SPH / MPH	Description
Vusb_hs_dma_mph_amba_apb.vhd1	MPH	Structure – BVCI DMA Device Controller Structure – AMBA-AHB
Vusb_hs_arb_mph.vhd1	MPH	Port Arbiter
vusb_hs_greycode.vhd1	All	Grey Code Tables
vusb_hs_otg.vhd1	OTG	On-The-Go Controller
vusb_hs_pe_dev_sm.vhd1	DEV, OTG	Protocol Engine Device Logic
vusb_hs_pe_hst_sm.vhd1	OTG, SPH, MPH	Protocol Engine Host Logic
Vusb_hs_pe_datapath.vhd1	All	Protocol Engine Data-path
Vusb_hs_pe_timebase.vhd1	All	Protocol Engine Time base
vusb_hs_pe_dev.vhd1	DEV	Protocol Engine Structure
vusb_hs_pe_otg.vhd1	OTG	Protocol Engine Structure
Vusb_hs_pe_sph.vhd1	SPH	Protocol Engine Structure
Vusb_hs_pe_mph.vhd1	MPH	Protocol Engine Structure
Vusb_hs_pe_tt.vhd1	MPH	Protocol Engine Structure
vusb_hs_pkg.vhd1	All	USB-HS Constant definitions
vusb_hs_portctrl.vhd1	All	Port Control Top Level
vusb_hs_portctrl_sm.vhd1	All	Port control state machine
vusb_hs_portctrl_rm.vhd1	All	Rate Matching Logic
vusb_hs_portctrl_rm_fifo.vhd1	All	Rate Matching Logic FIFO
vusb_hs_portctrl_sie.vhd1	All	Full speed serial interface
vusb_hs_portctrl_utmi.vhd1	All	UTMI characterization logic
vusb_hs_portctrl_ulpi.vhd1	All	ULPI
vusb_hs_rx_buf.vhd1	All	Receive Buffer Logic
vusb_hs_tt_rx_buf.vhd1	MPH	TT-Receive Buffer Logic
vusb_hs_tx_buf.vhd1	All	Transmit Buffer Logic
vusb_hs_tt_tx_buf.vhd1	MPH	TT-Transmit Buffer Logic
vusb_hs_up_int_bvci.vhd1	All	Microprocessor Interface Logic – BVCI
vusb_hs_up_int_amba_apb.vhd1	All	Microprocessor Interface Logic – AMBA-AHB
vusb_hs_up_int_busif_bvci.vhd1	All	Microprocessor Interface Bus Logic
vusb_hs_up_int_busif_amba_apb.vhd1	All	Microprocessor Interface Bus Logic
vusb_hs_up_int_regfile.vhd1	All	Microprocessor Interface Register File

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3.4.5 BEHAVIORAL_V or BEHAVIORAL_VHDL

These folder contains the VHDL or verilog models of the responder, phy and bus resolution. It also includes some testbench structural code.

3.5 Synthesis Notes

The following warnings are present in the Synopsys synthesis flow version 2003.06-SP1 and are noted here to indicate that these warnings have been thoroughly reviewed by the design team and have been deemed acceptable as justified here.

3.5.1 VHDL Compiler:

Warning: VHDL-93 generates different concatenation results from VHDL-87 in routine xxx line +++ in file '----.vhd1'. (VHDL-2285)

Justification: Synopsys reports this warning in circumstances where it is not really a problem. The coding standard that was applied prevents writing code that could cause this to be a real VHDL-87/93 mismatch.

3.5.2 HDL Compiler: (verilog)

Warning: Verilog 'integer' types are interpreted as signed; synthesized result will not match earlier versions of HDL Compiler. (VER-314)

Justification: The use of integers has been carefully considered in this design and they are designated for use in configuration parameters only. Therefore, they will not exhibit signed behavior. This warning is irrelevant.

3.5.3 General HDL:

Warning: The statement in routine ---- line +++ in 'xxxx.vhd1', is never reached. (HDL-395)

Warning: xxxx.v:++++: DEFAULT branch of CASE statement cannot be reached. (ELAB-311)

Warning: xxxx.v:++++: Case branches N to M are unreachable. (ELAB-314)

Justification: The vusb_hs_core_X is configured with constants and depending on the configuration selected, there will be code that is unexercised. These warnings are unavoidable by design.

Warning: xxxx.v:++++: Potential simulation-synthesis mismatch if index exceeds size of array 'signal_name'. (ELAB-349)

Justification: This behavior is as intended and will by design never exceed the bounds of the array at a time when the indexed array result is useful.

3.5.4 Test Compiler:

Warning: Data pin **inside** of cell tx_buf_clk_b (**out_port**) is driven by a clock/enable signal. (TEST-131)

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Justification: The vusb_hs_core[h] connects the proper clocks to the memory interface as determined by the clock configuration constant. Though upward paths for clocks may be undesirable, this connection is provided for ease of integration. It is possible to leave this output open or remove the port if the clock path to the memory is provided by the user. In either case, this warning is harmless.